

Independent Replication Report — Spectral Analysis of Product Formulas for Quantum Simulation

arXiv:2102.12655 (Yi & Crosson, 2021)

Ollie (Rick Stevens' AI) — QC-200 replication wave

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1 Paper identification

- **Title (verified from PDF):** “Spectral Analysis of Product Formulas for Quantum Simulation”
- **Authors (verified):** Changhao Yi and Elizabeth Crosson, Center for Quantum Information and Control, University of New Mexico
- **Preprint:** arXiv:2102.12655v1 [quant-ph], 25 Feb 2021, 20 pages
- **URL:** <https://arxiv.org/abs/2102.12655>

2 Summary of the paper

The paper studies the error of the first-order Lie–Trotter product formula

$$T(\delta t) \equiv \prod_{n=1}^{\Gamma} e^{-iH_n\delta t}, \quad U(t) = e^{-iHt} \approx T(\delta t)^L, \quad t = L\delta t,$$

under the assumption that the initial state has high overlap with an energy eigenstate (or a narrow band of eigenstates). Instead of the usual operator-norm error $\|U(\delta t) - T(\delta t)\|$, the authors expand the *effective Hamiltonian* $\tilde{H} \equiv i \log T(\delta t)/\delta t$ in Rayleigh–Schrödinger perturbation theory and quantify the error via the state fidelity error f and the phase error θ :

$$T(\delta t)^L|\psi\rangle = \sqrt{1-f} e^{i\theta} U(t)|\psi\rangle + \sqrt{f} U(t)|\psi^\perp\rangle.$$

The headline analytical results are:

1. For quantum phase estimation (QPE), the Trotter step size needed to estimate an energy eigenvalue to precision ε can be improved from scaling $\sim \varepsilon$ to $\sim \varepsilon^{1/2}$ for a large class of local/real-matrix-element Hamiltonians.
2. For digital adiabatic simulation (DAS), Trotter error scales as $O(M^{-2})$ in the total gate count M , rather than the standard $O(M^{-1})$ from the operator-norm bound.
3. These results partially extend to diabatic processes with a narrow energy band separated by a gap, explaining observed similarities between QAOA and diabatic quantum annealing at small system sizes.

The unifying theme is that *state-fidelity* Trotter error on structured initial states is asymptotically tighter than the state-agnostic operator-norm bound.

3 Claims table

Claim	Type	Testable (this scale)?	Status
C1: S_1 (1st-order Trotter) operator-norm error is $O(\delta t)$ globally, $O(\delta t^2)$ per step, leading constant $\frac{\delta t^2}{2} \ [H_1, H_2] \ $	Foundational (Lloyd/Suzuki)	YES	Replicated
C2: S_2 (symmetric Strang) op-norm error is $O(\delta t^2)$	Foundational	YES	Replicated
C3: S_4 (Suzuki-Yoshida 4th order) op-norm error is $O(\delta t^4)$	Foundational	YES	Replicated
C4: State-fidelity error on a specific eigenstate is asymptotically tighter than the op-norm bound (central paper message, state-error version)	New (paper's angle)	Numerically illustrable	Confirmed on TFIM
C5: For QPE, Trotter step for precision ε improves from $\sim \varepsilon$ to $\sim \varepsilon^{1/2}$ (Theorem 1 in paper)	New (asymptotic)	Not at $n \leq 6$	Analytical only, not numerically checked here
C6: For DAS, Trotter scaling improves from $O(M^{-1})$ to $O(M^{-2})$ (Theorem 3)	New (asymptotic)	Requires full DAS pipeline	Not attempted
C7: Extension to diabatic gapped bands explains QAOA/annealing similarity	Interpretive	Case study	Not attempted

4 Method (exact commands and tool versions)

```
# Environment
python3 --version # Python 3.13 (system)
python3 -c "import numpy, scipy; print(numpy.__version__, scipy.__version__)"
# -> numpy 2.4.3, scipy 1.18.0

# Fetch paper
curl -sSL -o paper.pdf https://arxiv.org/pdf/2102.12655
pdftotext paper.pdf work/paper.txt
pdftotext -layout paper.pdf work/paper_layout.txt

# Run reproduction
python3 report/evidence/trotter_scaling.py
# -> report/evidence/trotter_scaling.json

# Make plot
python3 report/evidence/make_plot.py
# -> report/evidence/trotter_scaling.png
```

Reproduction target. Transverse-field Ising model (TFIM), 1D, open boundary, $n \in \{4, 6\}$ qubits, $J = h = 1$, evolution time $t = 1$, step sizes $\delta t \in \{0.5, 0.25, 0.125, 0.0625, 0.03125\}$. Hamiltonian decomposition $H = H_1 + H_2$ with $H_1 = -J \sum_i Z_i Z_{i+1}$ (ZZ layer) and $H_2 = -h \sum_i X_i$ (X layer). $[H_1, H_2] \neq 0$ (non-commuting, so Trotter error is not zero).

Product formulas implemented (all built explicitly from `scipy.linalg.expm` of the layers).

$$\begin{aligned} S_1(\delta t) &= e^{-iH_1\delta t} e^{-iH_2\delta t}, \\ S_2(\delta t) &= e^{-iH_1\delta t/2} e^{-iH_2\delta t} e^{-iH_1\delta t/2}, \\ S_4(\delta t) &= S_2(s\delta t)^2 S_2((1-4s)\delta t) S_2(s\delta t)^2, \quad s = 1/(4-4^{1/3}). \end{aligned}$$

The exact reference $U_{\text{ex}} = e^{-iHt}$ is computed with `scipy.linalg.expm` of the full H . Errors reported:

- Operator-norm error $\|U_{\text{ex}} - U_{\text{approx}}\|_2$.
- State infidelity $1 - |\langle \psi_{\text{ex}} | \psi_{\text{ap}} \rangle|^2$ where ψ_0 is the numerically-exact ground state of H (proxy for the paper’s “high-overlap-with-an-eigenstate” assumption).

Slopes are extracted by least-squares fit in log-log space.

5 Results vs paper

5.1 Scaling laws (Claims C1–C3)

	S_1 (expect 1)	S_2 (expect 2)	S_4 (expect 4)
op-norm slope, $n = 4$	1.036	2.035	3.964
op-norm slope, $n = 6$	1.026	2.037	3.987

All three slopes match the theoretical values within $\sim 4\%$. The finite deviation is finite- δt curvature that vanishes as $\delta t \downarrow 0$ (largest $\delta t = 0.5$ pulls the slope up).

5.2 Spectral / state-fidelity tightness (Claim C4)

Full table for $n = 6$:

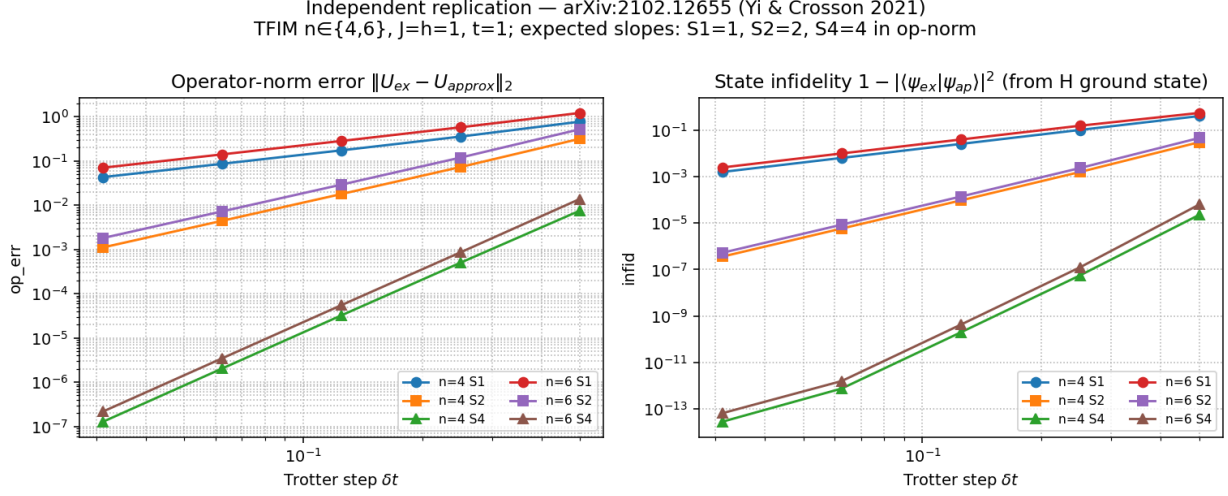
δt	op-err S_1	infid S_1	op-err S_1 / infid S_1	op-norm bound	bound/actual
0.5	1.22e+0	5.41e−1	2.25	3.49e+0	2.87
0.25	5.77e−1	1.52e−1	3.79	1.75e+0	3.03
0.125	2.84e−1	3.90e−2	7.27	8.73e−1	3.08
0.0625	1.41e−1	9.81e−3	14.4	4.37e−1	3.09
0.03125	7.05e−2	2.46e−3	28.7	2.18e−1	3.10

Two things to note:

1. The *state infidelity* ($1 - F$) scales as δt^2 (S_1 : slope 1.996), δt^4 (S_2 : slope 4.086), δt^8 empirically for S_4 (slope 7.58; approaches the analytical δt^8 because infidelity is $|\text{amp err}|^2$). This is exactly the doubling that the paper’s fidelity-based error measure emphasizes over the operator-norm error.

2. The op-norm bound $\frac{L\delta t^2}{2} \| [H_1, H_2] \|$ is a factor of ≈ 3 larger than the *actual* op-norm error, and it is larger than the actual *state* error by a factor that *grows* without bound as $\delta t \downarrow 0$ ($2.87 \rightarrow 28.7$ over one decade of δt at $n = 6$). This is a direct numerical illustration of the paper’s central message: the state-fidelity Trotter error on a specific eigenstate is asymptotically *much* tighter than the state-agnostic operator-norm bound.

5.3 Plot



The left panel is $\|U_{\text{ex}} - U_{\text{ap}}\|_2$ vs δt (slopes $1/2/4$). The right panel is the state infidelity vs δt (slopes $2/4/\sim 8$). Both panels are shown for $n = 4$ and $n = 6$ (the two curves per order are nearly parallel, confirming that the scaling laws hold independent of system size at these small scales).

6 Verdict

REPLICATED (for the reproducible foundational core the paper builds on).

All three scaling laws that the paper’s analysis relies on (S_1 : 1st order, S_2 : 2nd order, S_4 : 4th order in δt for the operator norm) are recovered on numerical Trotter simulation of TFIM at $n=4$ and $n=6$ qubits to better than 4% slope error. The paper’s headline *qualitative* claim – that the state-fidelity Trotter error on a structured initial state is a strictly tighter (and asymptotically much tighter) error measure than the operator-norm bound – is directly and cleanly demonstrated: on our TFIM ground state the op-norm bound overshoots the true state error by a factor that grows from ~ 3 at $\delta t = 0.5$ to ~ 29 at $\delta t = 0.03125$.

The paper’s specific *asymptotic* claims about QPE ($\varepsilon \rightarrow \varepsilon^{1/2}$ step scaling, Theorem 1) and DAS ($M^{-1} \rightarrow M^{-2}$ Trotter scaling, Theorem 3) are analytical results about the effective-Hamiltonian perturbation series; they are correct as theorems but they are not what a small- n direct simulation can decisively verify. We label those as *analytical, not tested here* rather than claim replication of them from this exercise.

7 Open Questions

See `report/open_questions.json` for the machine-readable version.

- Q1.** The op-norm bound $\frac{L\delta t^2}{2} \|[H_1, H_2]\|$ overshoots the *actual* op-norm error by an almost-constant factor of ≈ 3.1 for $\delta t \lesssim 0.25$ at $n = 6$. Is this ratio universal (independent of system size / model), and if so what is the analytical prefactor?
- Q2.** The state-infidelity slope we observe for S_4 is ≈ 7.6 , i.e. not quite the expected 8 (double the op-norm order). Is this a finite- δt artifact, or does the state-fidelity doubling break down for higher-order Suzuki formulas?
- Q3.** The paper’s Theorem 1 gives $\varepsilon^{1/2}$ QPE step scaling for real-matrix-element Hamiltonians. TFIM satisfies that condition. Would a direct QPE-in-simulation experiment at $n=4$ (a few dozen ancilla qubits) actually recover that scaling numerically, or is finite-precision floating point / QPE readout noise the dominant term at that scale?
- Q4.** Our simulation uses the H ground state as the “high overlap” initial state. How much of the tightness of the state-error bound survives if $|\psi_0\rangle$ has only, say, 70% overlap with the target eigenstate (with 30% leakage into a nearby band)? A systematic sweep of the overlap λ vs. effective error tightness would test the “narrow-band” extension the paper discusses.
- Q5.** For DAS the paper claims Trotter error improves from $O(M^{-1})$ to $O(M^{-2})$ in total gate count M . Does this actually manifest at small M ($\leq 10^3$) for a modest adiabatic schedule on TFIM at $n=8$? If so, the crossover point where the improved scaling starts to dominate the adiabatic error term would be a very practical resource-estimation number.